

Practical 4 – The Relay and the Protection Diode

Booklet 3, section 12, page 28. Success criteria 5.3, 5.4 and 5.5. Yenka only.

Everything you write goes in your jotter. Copy each numbered table before you start that section, and answer each question by its number.

The idea being tested

A relay is a switch worked by an electromagnet. A small current through its coil closes a separate set of contacts, so a small electronic circuit can switch a much larger one.

When the coil is switched off it produces a voltage spike that can damage the transistor. A protection diode across the coil, connected in reverse, stops it.

4.1 What you are testing

Aim: to switch a motor on a separate supply through a relay, and to find out what the protection diode does.

4.2 Predict

Table 4.2 – what switches on?

Light on the LDR	Transistor (on / off)	Relay coil (on / off)	Motor (on / off)
bright			
dark			

4.3 Simulate in Yenka

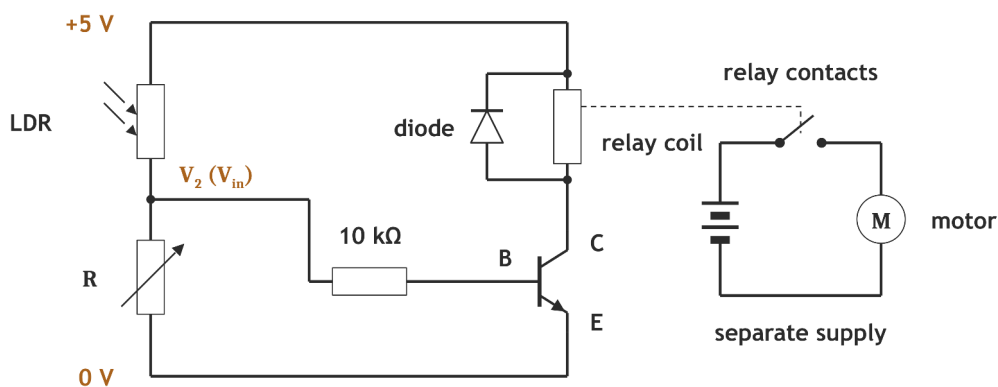


Table 4.3a – simulated readings

Light on the LDR	Base-emitter voltage (V)	Relay (on / off)	Motor (on / off)
bright			
dark			

Table 4.3b – the protection diode, in bright light

Circuit	Relay (on / off)	Motor (on / off)
diode the right way round		
diode turned round		
diode taken out		

Q1 Describe what happened when the diode was turned round, and suggest why. [2]

4.4 Compare

Q2 Compare Table 4.3a with your predictions in Table 4.2. State whether they matched and, if not, which row was different. [1]

4.5 What the results tell you

Q3 With the diode taken out, Yenka showed a component being destroyed when the relay switched off. Explain why, and how the diode prevents it. [2]

Q4 State the function of the relay in this circuit. [1]

Q5 Explain why a relay is used when a low-voltage transistor circuit has to switch a 230 V mains motor. [2]

Q6 Explain why the protection diode is connected in reverse across the coil. [2]

Q7 Describe the operation of the circuit as the light level falls. Refer to the resistance of the LDR, V_2 (V_{in}), the transistor, the relay and the motor. [5]

Q8 Draw a sub-system diagram for this circuit. Each output device needs a driver. [3]

Q9 Give one advantage of testing this circuit in simulation before building it. [1]

Q10 The motor drives a cooling fan that must switch on when a machine gets too hot. State which component replaces the LDR, and whether it goes at the top or the bottom of the voltage divider. [2]

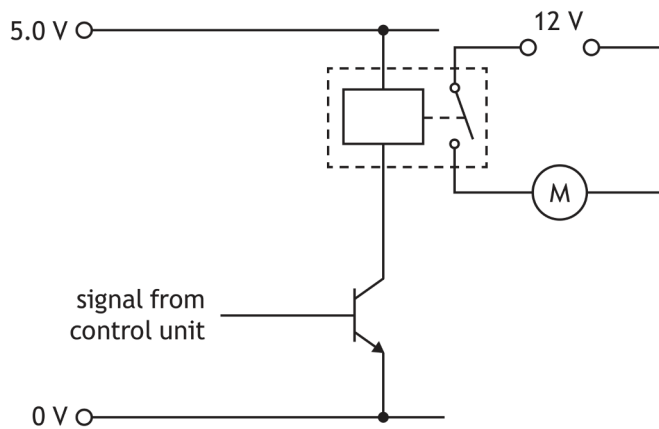
Make that change in Yenka and test your circuit against the specification.

Table 4.5 – does the fan meet the specification?

Specification point	Met? (yes / no)	Justification
i The fan switches on when the machine gets too hot.		
ii The fan switches off when the machine cools down.		
iii The fan runs from a supply separate from the control circuit.		

Q11 Copy Table 4.5 and complete it from your Yenka test. [3]

From a past exam paper



This is the circuit diagram for an output driver and motor.

Sources & credits: © Qualifications Scotland (SQA). National 5 Engineering Science 2026, Question 11(e).

Q12 Copy the circuit diagram and complete it by adding the symbol for a diode connected to protect the transistor. [3]